

BicoccApp

REST API Documentation

University of Milano-Bicocca Mobile Application

Base URL: `https://backoffice-app.unimib.it`

API Version: `v1`

Authentication: OpenID Connect + Token Headers

Content Type: `application/json`

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Authentication Overview

The BicoccApp API uses a two-step authentication process. First, the app initiates an OpenID Connect flow with the university's identity provider. After successful authentication, the API returns three headers that must be included in all subsequent authenticated requests:

Header	Description
uid	User identifier (email address)
client	Client session token
access-token	Access token for API authorization

Authentication

Endpoints for user authentication using OpenID Connect protocol with the university's identity provider.

GET /api/v1/auth/openid_connect

Initiates the OpenID Connect authentication flow. Redirects the user to the university's identity provider (IdP) for login.

Authentication:	Not required
Response Status:	302 Redirect

GET /api/v1/auth/openid_connect/callback

Handles the callback from the identity provider after successful authentication. Returns authentication tokens.

Authentication:	Not required
Query Parameters:	code, state
Response Status:	302 Redirect

Example Request:

```
GET /api/v1/auth/openid_connect/callback?code=AAadzZWnyZXQxRgI_K9wiwbkRri--dufAFwaXYEIr-W0iStAZ3XtldXUlnNenJEyIWqnu7fD-GUiC9VQCXjpd5jJmnUPdTKKWTbAlW91ZrshwuzkwfRaQf3zx5zV8kS_uwYlYsoj20F1gDja06jtZYfZV4FY56DHN6HkJ97RNMkIG7eCSvWP-yreg6N7nPgkWXIWu5st1G5a1Bhw_m8oyTqmedqnowb9D4qN0-qNIRyzpWYHvs9xus3Br0YPUCVo519LEjd2k6xGv1_cqswwnSLCBE35c7MX0pk0THXM8MmWLDa6QNo8uxWQpzSdxJtZCPtM-a1NtIZdQ3ecuSZz_VAdqtbE7_gZlH2jdFVxgRiIgZaPsSqY2lxy_o0Mmvsh-JsV1Xu02EtRg0p0gH1FeAcx4eC9diftr1-w0VkZ4QKp5ogRySDv9mS4EtL-MCMol3Uln6uYQi3Cu4DdTIwJ5kgwG9ZneZYyq73EkaU1B54Z_sRd5zjtaRfLae-dplh_8MrzSR2z40ALD_Lz8cGyGdbDycxQfPO-gHTC8MtwxIlhuMwYgfaF0fQbextWb4oFptse8sgqCPcWRdsxLkZ6hEzAd0pVLK1tjMFibnkPyBaQntDnRGzBZ6LPDcop8DF0LFyTArknfey2Uftq7akUCAwC0KqvmZetPntpjqwJkQ3NdKff8TxU0BEqjJRGY51PCMjTaKHSqCqKgN0-Av5LsBCtDiYeBPptvry3c0ZGXJAK7Hhi6K&state;=57ac632df2c337110f11e8b328ffab32
```

User Profile & Career

Endpoints for retrieving user profile information, academic career details, exam history, fees, and registrations.

Profile

GET /api/v1/user_profile

Retrieves the authenticated user's profile information including personal details and associated academic careers.

Authentication:	Required (uid, client, access-token headers)
Query Parameters:	fiscalCode
Response Status:	200 OK

Example Request:

```
GET /api/v1/user_profile?fiscalCode=TRALSN05A12F205A
```

Response Example:

```
{ "user": { "appuser_id": 16474, "personId": 454914, "userId": "a.autiero2", "fiscalCode": "TRALSN05A12F205A",  
"name": "ALESSANDRO AUTIERO", "surname": "", "mobile": "+39 3495089819", "email": "a.autiero2@campus.unimib.it",  
"fotoId": 349766, "photo": null, "has_calendar": true }, "careers": [ { "matricId": 438589, "selected": true,  
"oldCode": false, "matricCode": "909697", "studentId": 6433373, "typeCourseC ...
```

Career & Academics

GET /api/v1/user_career

Fetches detailed information about the student's academic career including averages, statistics, and degree program details.

Authentication:	Required (uid, client, access-token headers)
Query Parameters:	matricId, personId, studentId, typeTitleCode
Response Status:	200 OK

Example Request:

```
GET /api/v1/user_career?matricId=438589&personId;=454914&studentId;=6433373&typeTitleCode;=L2
```

Response Example:

```
{ "career": { "averages": [ { "base": 30, "basedefinition": "CDSORD", "weighted": 27.33, "arithmetic": 27.33 },  
{ "base": 110, "basedefinition": "CDS", "weighted": 100.22, "arithmetic": 100.22 } ], "stats": { "valueCode":  
"C", "valueDescr": "Crediti", "examsDone": 14, "totalToDo": 180.0, "totalDone": 99.0 }, "degree": { "score":  
"not available", ...
```

GET /api/v1/user_exams

Retrieves the history of exams taken by the student, including grades and dates.

Authentication:	Required (uid, client, access-token headers)
Query Parameters:	matricId
Response Status:	200 OK

Example Request:

GET /api/v1/user_exams?matricId=438589

Response Example:

```
{ "career": { "notations": [ { "year": 2023, "dateExam": "23/01/2024 00:00:00", "laudFlag": 0, "grade": 24.0 }, { "year": 2023, "dateExam": "02/02/2024 00:00:00", "laudFlag": 0, "grade": 30.0 }, { "year": 2023, "dateExam": "13/02/2024 00:00:00", "laudFlag": 0, "grade": 25.0 }, { "year": 2023, "dateExam": "18/06/2024 00:00:00", "laudFlag": 0, "grade": 27.0 }, { "year": 2023, "dateExam": "28/06/2024 00:00:00", "laudFlag": 0, "grade": 29.0 }, { "year": 2023, "dateExam": "15/07/2024 00:00:00", "laudFlag": 1, "grade": 30.0 }, { "year": 2023, "dateExam": "13/09/2024 00:00:00", "laudFlag": 0, "grade": 24.0 } ] }
```

GET /api/v1/user_appeals

Gets information about exam sessions the student is registered for or can register to.

Authentication:	Required (uid, client, access-token headers)
Query Parameters:	matricId, personId
Response Status:	200 OK

Example Request:

GET /api/v1/user_appeals?matricId=438589&personId=454914

Response Example:

```
{ "career": { "courses": [ { "yearFreqId": 2024, "activityCode": "E3101Q103", "activityItemId": 10657399, "activityDescr": "BASI DI DATI", "courseYear": 2, "activityId_o": 37792, "cdsId_o": 10161, "activityId": 37792, "cdsId": 10161, "appeals": [ { "appealId": 447866, "activityAppealId": 180, "activityCode": "E3101Q103", "activityDescr": "BASI DI DATI", "activityId": 37792, "cdsId": 10161, "yearCalendaId": 2024, "reserved": false, "bookable": false, "registrationStartDate": "21/12/2025 00:00:00", "registrationEndDate": "17/01/2026 00:00:00", "..." } ] } ] }
```

Fees & Registrations

GET /api/v1/user_fees

Retrieves the student's tuition fee payment history and pending payments.

Authentication:	Required (uid, client, access-token headers)
Query Parameters:	personId, matricId
Response Status:	200 OK

Example Request:

GET /api/v1/user_fees?personId=454914&matricId=438589

Response Example:

```
{ "career": { "fees": [ { "yearId": 2025, "fiscalCode": "TRALSN05A12F205A", "invoiceId": 3023477, "paymentId": 3023477, "paidFlag": 1, "description": "PENALE PAGAMENTO TARDIVO 1A RATA DI 3 - A.A. 2025/2026", "releaseDate": null, "paymentDate": "04/10/2025", "mavDescription": "PENALE PAGAMENTO TARDIVO 1A RATA DI 3", "mavDescriptionNext": null, "invoiceCanceled": 0, "invoiceWrong": null, "invoiceArrearId": 3013623, "invoiceOverdue": null, "invoiceAmount": "15.60", "invoiceAmountPaid": "15.60", "cashBy": "PA", "arrearAddFlag": 0, "arrearAddDate": null, "..." } ] }
```

GET /api/v1/user_registrations

Gets information about the student's course registrations.

Authentication: Required (uid, client, access-token headers)
Query Parameters: matricId
Response Status: 200 OK

Example Request:

GET /api/v1/user_registrations?matricId=438589

Response Example:

```
{ "career": { "registrations": [] } }
```

Calendar & Schedule

Endpoints for managing the academic calendar, viewing scheduled lessons and exams, and adding courses to the personal calendar.

GET /api/v1/calendar

Retrieves the academic calendar with scheduled events, lessons, and exams for a specific date or date range.

Authentication:	Required (uid, client, access-token headers)
Query Parameters:	matricId, date
Response Status:	200 OK

Example Request:

```
GET /api/v1/calendar?matricId=438589&date;=2025-11-27
```

Response Example:

```
{ "calendar": [ { "day": "27-11-2025", "events": [ { "eventId": "1271370", "courseName": "PROGRAMMAZIONE C++",  
"courseCode": "509307", "partition": "Lezione,Esame", "date": "27-11-2025", "time": "08:30 - 11:30", "type":  
"Lesson", "day": "Thursday", "roomCode": "U14-LAB14A1", "room": "LAB14A1 (U14)", "note": null, "canceled": "0",  
"maps": null ...
```

GET /api/v1/calendar_courses

Lists all courses available for adding to the personal calendar.

Authentication:	Required (uid, client, access-token headers)
Response Status:	200 OK

Response Example:

```
{ "courses": [ { "cdsCode": "E3101Q", "activityCode": "E3101Q132", "lessonName": "ALGEBRA LINEARE E GEOMETRIA", "partitio  
n": "Exam", "courseCode": "ins_E3101Q_E3101Q132_1_GGG_T1_BORGH", "matricId": null, "activityItemId": null, "is_active":  
false, "is_booklet": false }, { "cdsCode": "E3101Q", "activityCode": "E3101Q132", "lessonName": "ALGEBRA LINEARE E  
GEOMETRIA - T1 - ESE (gr. A)", "partition": "Lesson", "courseCode": "E3101Q_E3101Q132_1_GGG_E3101Q132M_T1GEA_ESE", "  
matricId": null, "activityItemId": null, "is_active": true, "is_book...
```

POST /api/v1/set_calendar

Adds a course to the user's personal calendar for tracking lessons and exams.

Authentication:	Required (uid, client, access-token headers)
Request Body:	application/x-www-form-urlencoded
Response Status:	200 OK

Request Body Example:

```
{ "user_id": "16474", "cdsCode": "E3101Q", "activityCode": "E3101Q113", "lessonName": "ANALISI E PROGETTO  
DI ALGORITMI (T2GLA LAB TURNO 2) (T2)", "courseCode": "509297", "partition": "Lesson/Exam" }
```

Response Example:

```
{ "message": "Course added to calendar correctly", "status": 200 }
```

Courses & Lessons

Endpoints for browsing courses, viewing course details, and the course selection wizard.

Course Details

GET /api/v1/course_detail

Retrieves detailed information about a specific course including teachers, scheduled events, and room locations.

Authentication:	Required (uid, client, access-token headers)
Query Parameters:	activityCode, courseCode
Response Status:	200 OK

Example Request:

```
GET /api/v1/course_detail?activityCode=E3101Q128&courseCode;=509281
```

Response Example:

```
{ "activity_code": "E3101Q128", "cds_code": "E3101Q", "lesson_name": "RICERCA OPERATIVA E PIANIFICAZIONE DELLE RISORSE (T1 FRONTALE TURNO 1)", "partition": "Lesson,Exam", "course_code": "509281", "teachers": [ { "teacher_surname": "MESSINA", "teacher_email": "enza.messina@unimib.it", "teacher_code": "013421", "teacher_name": "VINCENZINA", "teacher_key": "enza.messina@unimib.it" } ], "events": [ { "eventId": "1316851", "courseName": "RICERCA OPERATIVA E PIANIFICAZIONE DELLE RISORSE (T1 FRONTALE TURNO 1)", "courseCode": "509281", "...
```

Course Wizard

GET /api/v1/wizard/categories

Gets the list of academic area categories (e.g., Economics, Law, Medicine, Science).

Authentication:	Required (uid, client, access-token headers)
Response Status:	200 OK

Response Example:

```
{ "categories": [ { "name": "Area Economico statistica", "code": "ECOSTAT" }, { "name": "Area Giuridica", "code": "GIUR" }, { "name": "Area Medica", "code": "MED" }, { "name": "Area Psicologica", "code": "PSICO" }, { "name": "Area Scientifica", "code": "SCIENT" }, { "name": "Area Scienze della Formazione", "code": "FORMA" }, { "name": "Area Sociologica", "code": ...
```

GET /api/v1/wizard/degrees

Retrieves available degree types for a specific academic category.

Authentication:	Required (uid, client, access-token headers)
Query Parameters:	category_code
Response Status:	200 OK

Example Request:

```
GET /api/v1/wizard/degrees?category_code=ECOSTAT
```

Response Example:


```
{ "degrees": [ { "name": "Corso di Laurea", "code": "L2" }, { "name": "Corso di Laurea Magistrale", "code": "LM" } ] }
```

GET /api/v1/wizard/courses

Lists degree programs available for a category and degree type combination.

Authentication:	Required (uid, client, access-token headers)
Query Parameters:	category_code, degree_code
Response Status:	200 OK

Example Request:

```
GET /api/v1/wizard/courses?category_code=ECOSTAT&degree_code=L2
```

Response Example:

```
{ "courses": [ { "name": "ECONOMIA AZIENDALE E MANAGEMENT", "code": "E1805M" }, { "name": "ECONOMIA DELLE  
BANCHE, DELLE ASSICURAZIONI E DEGLI INTERMEDIARI FINANZIARI", "code": "E1803M" }, { "name": "ECONOMIA E  
AMMINISTRAZIONE DELLE IMPRESE", "code": "E1802M-E1807M" }, { "name": "ECONOMIA, ANALISI DEI DATI E MANAGEMENT",  
"code": "E3303M-E3305M" }, { "name": "ECONOMIA, BANCA E FINANZA", "code": "E1808 ...
```

GET /api/v1/wizard/lessons

Gets the list of lessons/courses for a specific degree program.

Authentication:	Required (uid, client, access-token headers)
Query Parameters:	category_code, degree_code, course_code
Response Status:	200 OK

Example Request:

```
GET /api/v1/wizard/lessons?category_code=ECOSTAT&degree_code=L2&course_code=E1805M
```

Response Example:

```
{ "lessons": { "1": [ { "activity_code": "E1805M003", "cds_code": "E1805M", "lesson_name": "DESIGN, CRITICAL AND CREATIVE  
THINKING (TU FRONTALE TURNO UNICO) (TU)", "course_code": "510103", "partition": "Lesson/Exam", "teachers": [ { "teacher  
_surname": "MARAFIOTI", "teacher_email": "elisabetta.marafioti@unimib.it", "teacher_code": "002197", "teacher_name": "  
ELISABETTA", "teacher_id": 106480 } ] }, { "activity_code": "E1805M001", "cds_code": "E1805M", "lesson_name": "ECONOMIA  
AZIENDALE (TU)", "course_code": "510106", "partition": "Lesson/..."
```

GET /api/v1/wizard/user_courses

Retrieves the courses associated with the authenticated user's degree program.

Authentication:	Required (uid, client, access-token headers)
Query Parameters:	matricId
Response Status:	200 OK

Example Request:

```
GET /api/v1/wizard/user_courses?matricId=438589
```

Response Example:

```
{ "lessons": { "3": [ { "activity_code": "E3101Q113", "cds_code": "E3101Q", "lesson_name": "ANALISI E PROGETTO DI ALGORITMI  
(T2 FRONTALE TURNO 2 + T2GEA ESER TURNO 2) (T2)", "course_code": "509295", "partition": "Lesson/Exam", "teachers": [ {  
"teacher_surname": "BONIZZONI", "teacher_email": "paola.bonizzoni@unimib.it", "teacher_code": "012279", "teacher_name
```

```
":"PAOLA","teacher_id":104700}}},{ "activity_code":"E3101Q113", "cds_code":"E3101Q", "lesson_name":"ANALISI E  
PROGETTO DI ALGORITMI (T2GLA LAB TURNO 2) (T2)", "course_co...
```

Notifications & Messages

Endpoints for managing notifications, alerts, conversations with teachers, and appointment requests.

GET /api/v1/alerts

Fetches notifications and alerts for the user including exam results, communications, and announcements.

Authentication: Required (uid, client, access-token headers)
Response Status: 200 OK

Response Example:

```
{ "alerts_to_read": 30, "alerts": [ { "event_id": 342054, "alert_id": 154066, "title": "Pubblicazione Voto Appello", "date": "17-01-2024", "read": false, "rows": [ "Esito esame", "Gentile ALESSANDRO AUTIERO, ", "Le comunichiamo che l'esito relativo alla prova dell'attività didattica ANALISI MATEMATICA [E3101Q100], da lei sostenuta in data 15/01/2024, è stato pubblicato sulla sua bacheca esiti.", "Le ricordiamo che potrà accettare o meno l'esito assegnato fino al giorno 20/01/2024.", "Cordiali saluti,", "Il responsabile..." ] } ] }
```

GET /api/v1/conversations

Retrieves the list of message conversations the user has with teachers.

Authentication: Required (uid, client, access-token headers)
Query Parameters: page
Response Status: 200 OK

Example Request:

```
GET /api/v1/conversations?page=1
```

Response Example:

```
{ "conversations": [], "unread_messages": 0 }
```

GET /api/v1/messages/info_message

Gets system-wide informational messages to display in the app.

Authentication: Not required
Response Status: 200 OK

Response Example:

```
{}
```

POST /api/v1/messages/appointment

Sends an appointment request message to a teacher.

Authentication: Required (uid, client, access-token headers)
Request Body: application/x-www-form-urlencoded
Response Status: 403 Forbidden

Request Body Example:

teacher_key=enza.messina@unimib.it student_id=16474 message_body=Good morning Professor. Can you tell me when would you be available for a meeting? Thank you

Response Example:

```
{ "error": "Teacher not registered with BicoccApp" }
```

Campus Information

Endpoints for accessing campus maps, points of interest, and teacher contact information.

GET /api/v1/point_of_interests

Retrieves campus map data including buildings, cafeterias, libraries, and departments with coordinates.

Authentication: Required (uid, client, access-token headers)
Response Status: 200 OK

Response Example:

```
{ "maps": { "filters": [ { "type": "buildings", "color": "hueBlue" }, { "type": "caffeteria", "color": "hueYellow" }, { "type": "departments", "color": "hueRed" }, { "type": "library", "color": "hueAzure" } ], "map_locations": [ { "name": "Bar U3 (Bios)", "latitude": "45.5137", "longitude": "9.21210", "description": "Edificio U3 (Bios) - Piazza della Scienza, 2, 20126 Milano", "type": "caffeteria" }, { "name": "Biblioteca - Sede di Medicina", "latitude": "45.6040612158202", "longitude": "9.261045107557583", "description": "Edificio U8 (Asclepio..." }
```

GET /api/v1/teacher

Gets detailed information about a specific teacher including contact details, office hours, and location.

Authentication: Required (uid, client, access-token headers)
Query Parameters: teacherEmail
Response Status: 200 OK

Example Request:

```
GET /api/v1/teacher?teacherEmail=enza.messina@unimib.it
```

Response Example:

```
{ "teacher": { "title": "Professoressa", "name": "Vincenzina", "surname": "Messina", "email": "enza.messina@unimib.it", "email_alternate": "enza.messy@gmail.com", "teacherCode": "013421", "gender": "F", "teacherKey": "enza.messina@unimib.it", "receivesOn": "<p>Lunedì dalle 11:30 alle 13:30 su appuntamento</p>\n", "roles": [ { "roleDescr": "Professore/ssa ordinario/a" } ], "phones": [ { "officePhone": "0264487833" ... }
```

App Navigation

Endpoints for retrieving the app's navigation structure and menu configuration.

GET /api/v1/app_navigations/navigation_drawer

Retrieves the app's navigation drawer menu structure and configuration.

Authentication:	Required (uid, client, access-token headers)
Query Parameters:	locale
Response Status:	200 OK

Example Request:

GET /api/v1/app_navigations/navigation_drawer?locale=en

Response Example:

```
{
  "navigation_drawers": [
    {
      "id": 11,
      "title": "Notifications",
      "url": "NoticesScreen",
      "order": 1,
      "navigation_type": "drawer",
      "action_type": "app_navigation",
      "level": 1,
      "student_only": true,
      "public": false,
      "image": "",
      "submenus": []
    },
    {
      "id": 27,
      "title": "Student Services",
      "url": "linksScreen",
      "order": 2,
      "navigation_type": "drawer",
      "action_type": "app_navigation",
      "level": 1,
      "student_only": true,
      "public": false,
      "image": "",
      "submenus": [
        {
          "id": 62,
          "title": "Manage bookings",
          "url": "https://gestioneorari.didattica.unimib.it/portale..."
        }
      ]
    }
  ]
}
```